

Coolant VFD - Troubleshooting Guide - NGC

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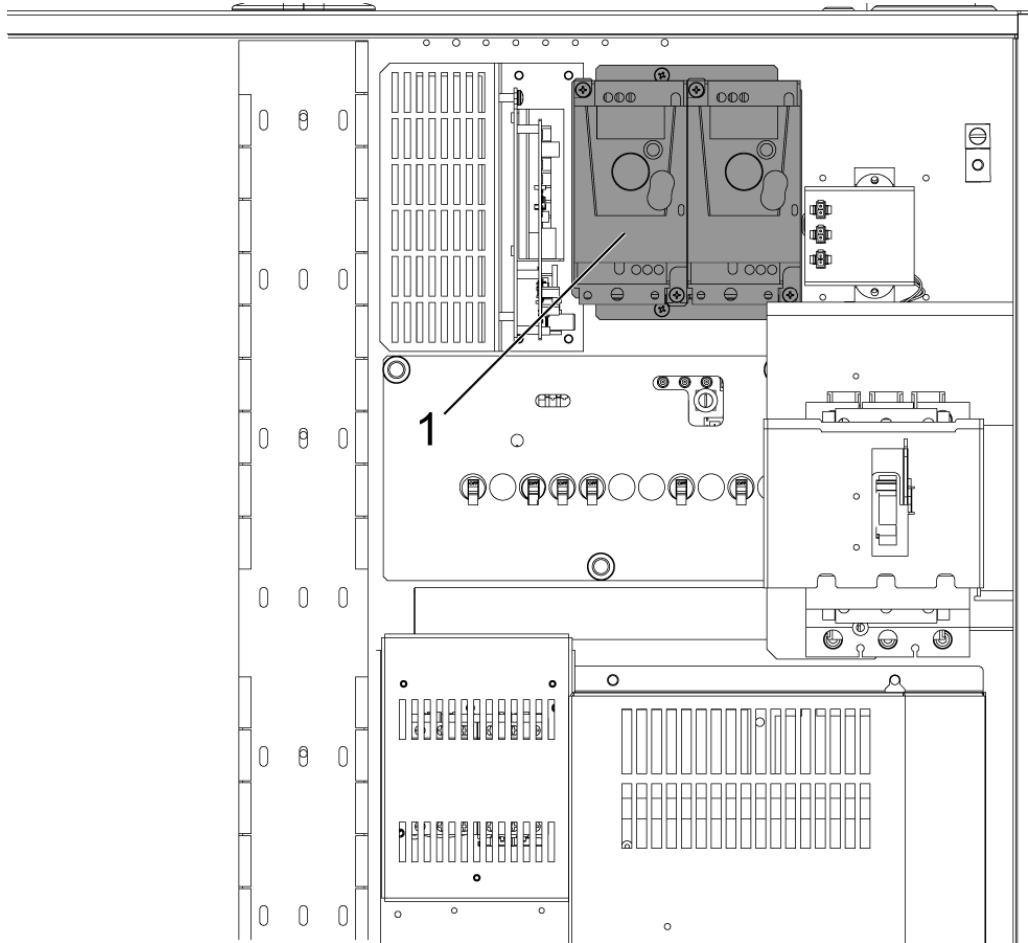
TG0056

Variable Frequency Drive (VFD) - Coolant Pumps

UMC's built after Nov 2019 and EC-1600's built after Jan 2020 now have a variable frequency drive (VFD) [1] units. These VFD units are located on the control cabinet above the PSUP PCB, these units will drive the coolant pump and the shower coolant pump (if equipped).

A VFD is a Variable Frequency Drive. VFD's run motors, and change their output frequency based on a command. This changes the speed of the motor. When used with coolant pump motors, coolant flow and pressure can be controlled, and not depend on the local line frequency of 50 or 60Hz. Some machines have up to two coolant pumps: standard and shower coolant; each is controlled by a separate VFD.

More troubleshooting information can be found on the manufacturer's user manual.



This feature allows you to select the desired pressure of the coolant pump.

An optional P-Code can now be specified along with an M08. This P-Code can be used to specify the desired pressure level of the coolant pump: P0 = Low Pressure P1 = Normal Pressure P2 = High Pressure **Note:** If no P-Code is specified, or the specified P-Code is out of range, then normal pressure will be used.

Note: Specifying a P2 (High Pressure), the coolant pump will operate at louder sound level, this is normal.

The following parameters enable the VFD drive:

NOTE: These parameters are set at the factory and listed here for reference only.

- 2232 Coolant VDF Type
- 2233 Shower Coolant VFD Type

When set to 0 = no VFD and 1= VFD.

Symptom Table

Symptom	Possible Cause	Corrective Action
Machine with SCHNEIDER Variable Frequency Drive - The coolant takes too long to flow out the nozzles or Pcool.	The variable frequency drive has an incorrect acceleration parameter value.	Make sure the acceleration parameter on the variable frequency drive (VFD) is set to 0.3 seconds. Refer to the VFD Acceleration Parameter Verification section below.

Alarm 9823 Coolant VFD Fault Alarm 9824 Shower Coolant VFD Fault	The coolant or shower pump is not plugged in.	Make sure the coolant or shower pump is connected to the coolant port. Check that the coolant cables are connected.
	The variable frequency drive reported a fault.	Open the control cabinet and see what fault was generated by the VFD. Refer to the VFD Manual, see Variable Frequency Drive (VFD) section below.
	The communication between the VFD unit and the control are not correct.	See Variable Frequency Drive (VFD) section below.
	The VFD overloaded while running at the highest pressure setting P2	See Variable Frequency Drive (VFD) section below for VFD overload troubleshooting process. Contact Haas Service to notify them of this issue.
	Motor overload "OFL" alarm on the VFD while using the washdown nozzle.	Check the coolant tank level and make sure the pump is not cavitating, and verify the coolant tank is free of chips.
The Coolant VFD is commanded to turn on (M08 PX), and the VFD shows "READY", and the pump does not turn on or takes several minutes to turn on.	The coolant VFD enable signal is not being read.	Determine if the drive has the Coolant VFD - Retrofit Kit Installed. Refer to the Coolant VFD - Retrofit Kit - AD0602 procedure.

Shortly after the replacement of a coolant VFD, intermittent alarm 9823 Coolant VFD Fault generated on Haas control, and error code "INF3" generated on the Schneider VFD. The machine is also built before 2/9/2022.	Incorrect VFD replacement kit used.	Order and install the correct VFD replacement kit. Follow the instructions in AD0602 and AD0516. For machines built before 2/9/2022 with VFD 69-0025. The machine must be upgraded with the following kits depending on the number of VFDs installed. • 93-3499 - SINGLE VFD RETROFIT KIT • 93-3498 - DUAL VFD RETROFIT KIT Important: Do not use VFD Adapter PCB 34-3031A on machines with I/O PCB boards 34-3490C as the VFD adapter is built into this board version.
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VFD I/O Signals

The commands in a program cause an enable and speed command signal from the IOPCB to the VFD. Power for the VFD is supplied directly from the Main Transformer.

Use the below table to make sure the CNC control is sending and receiving the correct signals from the Coolant VFD unit. **Note:** The value on the readout can be +/- 3Hz.

Command	Input / Output	Diagnostics Bits	Voltages at VFD (Schneider)	Voltages at VFD (Delta)	VFD DISPLAY

M08 P0	INPUT 112	VFD_COOLANT_FAULT = 0	RIA / COM ≈ 0VDC	RA / COM ≈ 0 VDC	49.9 Hz
	OUTPUT 170	VFD_COOLANT_BIT_1 = 1	AI1 / COM ≈ 6.24 VDC	AVI / COM ≈ 6.25 VDC	
	OUTPUT 171	VFD_COOLANT_BIT_2 = 0			
	OUTPUT 172	VFD_COOLANT_ENABLE = 1	LI1 / COM ≈ 0.3 VDC	MI1 / COM ≈ 0.95 VDC	
M08 P1	INPUT 112	VFD_COOLANT_FAULT = 0	RIA / COM ≈ 0 VDC	RA / COM ≈ 0 VDC	59.9 Hz
	OUTPUT 170	VFD_COOLANT_BIT_1 = 0	AI1 / COM ≈ 7.5 VDC	AVI / COM ≈ 7.5 VDC	
	OUTPUT 171	VFD_COOLANT_BIT_2 = 1			
	OUTPUT 172	VFD_COOLANT_ENABLE = 1	LI1 / COM ≈ 0.3 VDC	MI1 / COM ≈ 0.95 VDC	

M08 P2	INPUT 112	VFD_COOLANT_FAULT = 0	RIA / COM ≈ 0VDC	RA / COM ≈ 0 VDC	78.8 Hz
	OUTPUT 170	VFD_COOLANT_BIT_1 = 1	AI1 / COM ≈ 9.9 VDC	AVI / COM ≈ 9.9 VDC	
	OUTPUT 171	VFD_COOLANT_BIT_2 = 1			
	OUTPUT 172	VFD_COOLANT_ENABLE = 1	LI1 / COM ≈ 0.3 VDC	MI1 / COM ≈ 0.95 VDC	

Alarm	INPUT/OUTPUT	DIAGNOSTIC BITS	VOLTAGES AT VFD (SCHNEIDER)	VOLTAGES AT VFD (DELTA)
9823 COOLANT VFD FAULT	INPUT 112	VFD_COOLANT_FAULT = 1	RIA / COM ≈ 12 VDC	RA / COM ≈ 12VDC
	OUTPUT 170	VFD_COOLANT_BIT_1 = 0	AI1 / COM ≈ 0.0 VDC	AVI / COM ≈ 0.16 VDC
	OUTPUT 171	VFD_COOLANT_BIT_2 = 0		
	OUTPUT 172	VFD_COOLANT_ENABLE = 0	LI1 / COM ≈ 12 VDC	MI1 / COM ≈ 12 VDC

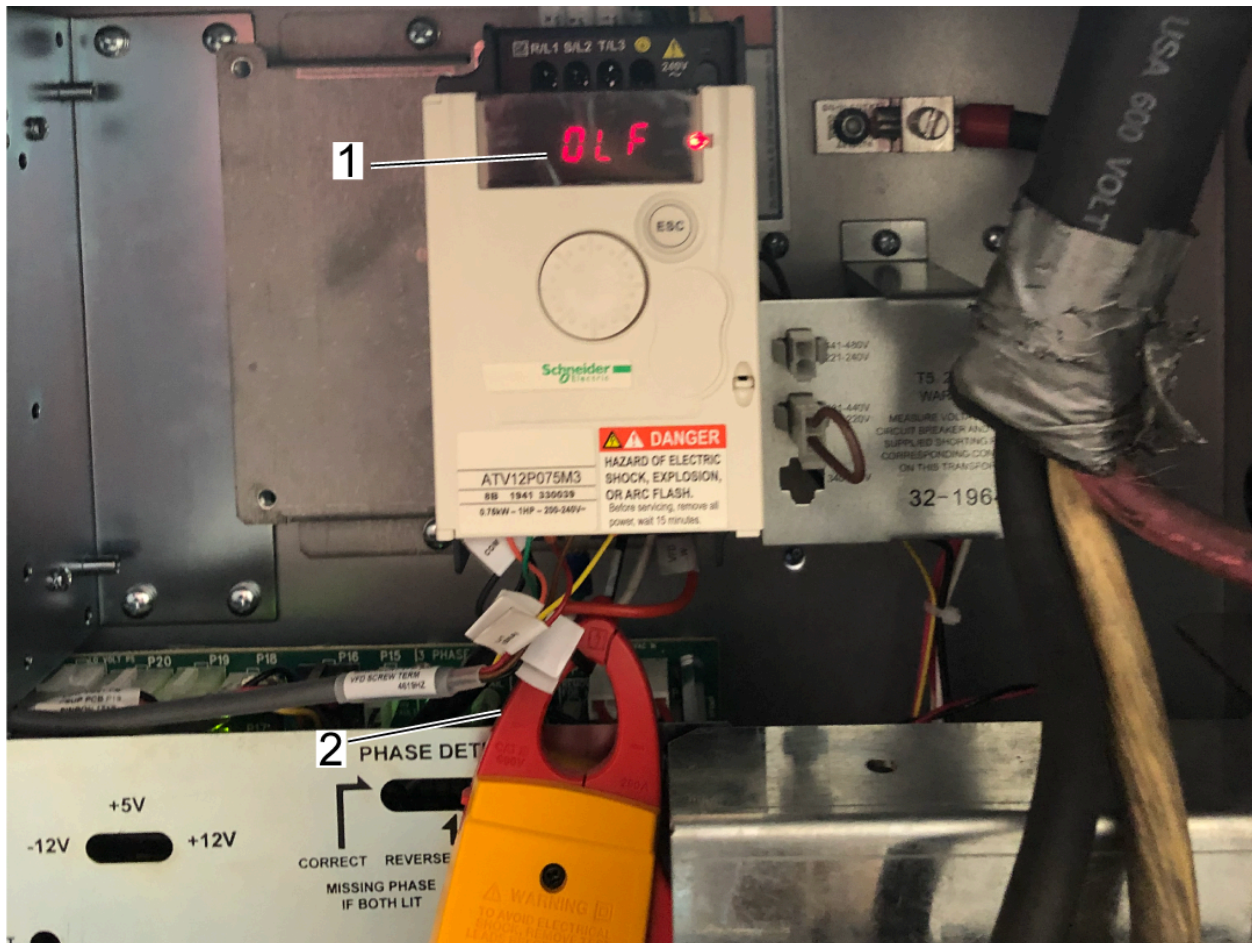
VFD Acceleration Parameter Verification

The following procedure will show you how to make sure the acceleration parameter on the Schneider ATV12 VFD is set correctly. With the machine power on. **WARNING:** Be careful not to touch the incoming High Voltage Power wires. If you don't feel comfortable with this procedure please contact your Haas Factory Outlet.

1. Press the dial button one time

2. Rotate the dial until you find **CONF** displayed on the screen.
3. Press the dial button one time.
4. Rotate the dial until you find **ACC** displayed on the screen.
5. Press the dial button one time.
6. Rotate the dial until the screen reads **0.3**
7. Press the dial button one time.
8. Press the **ESC** button until the screen reads **RDY**

VFD overload troubleshooting



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The VFD will need to be tested at each of the pressure settings at each of the pressure settings make sure they are within the range of the following.

NOTE: The frequency can be seen where the overload alarm is shown [1]. Use a clamp meter [2] to record the current at the different settings.

~50 Hz at P0, test current at this setting and record

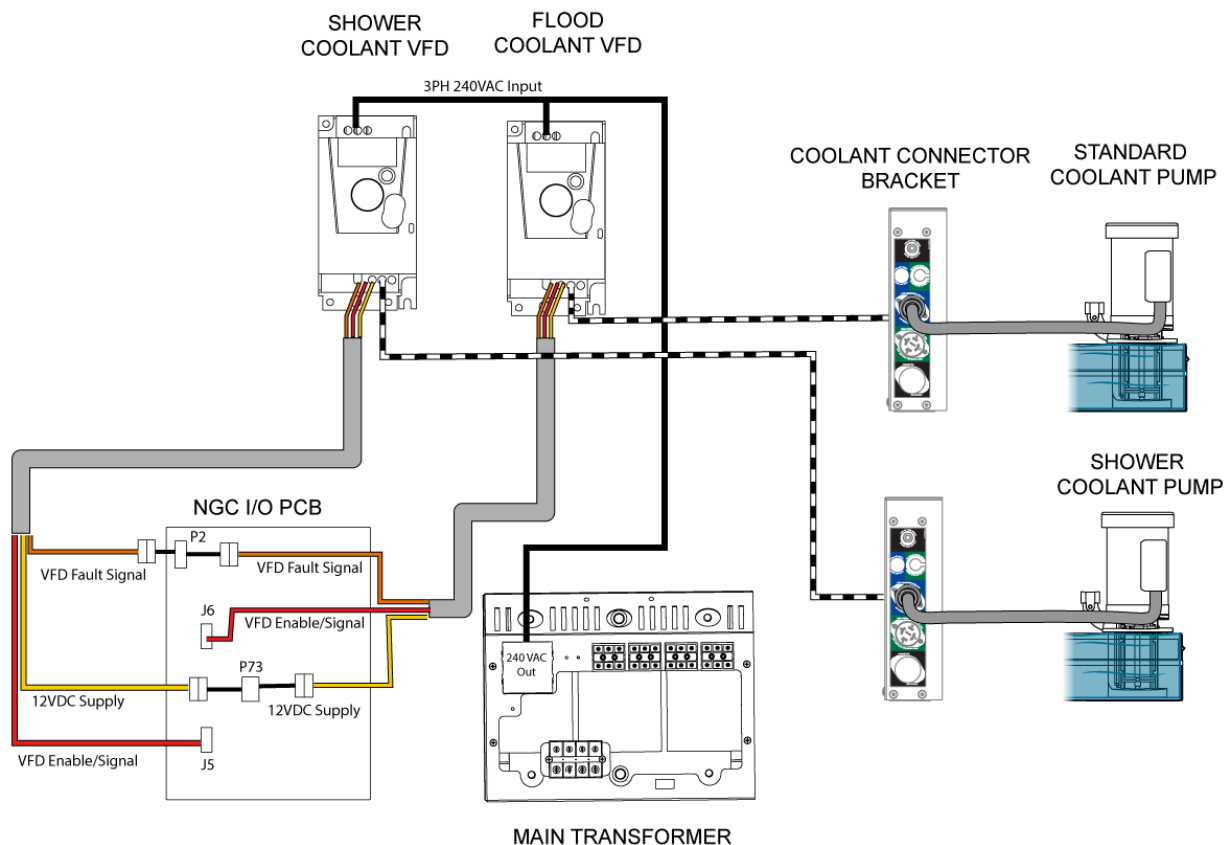
~60 Hz at P1, test current at this setting and record

~80 Hz at P2, test current at this setting and record.

NOTE: Frequency should be within +/- 3Hz. If you are having overload issues contact Haas service.

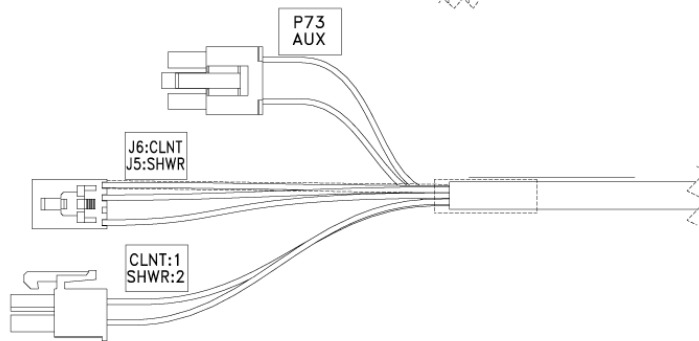
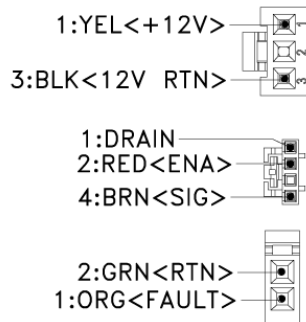
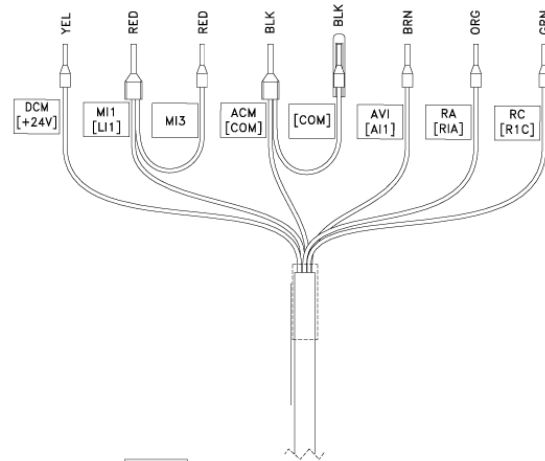
Electrical Diagrams

Use this electrical diagram for coolant pumps that are being driven by the Variable Frequency Drive With Cable 33-9934B.



33-9334B VFD Detail Cable Diagram

DELTA & SCHNEIDER VFD EQUIVALENT CONNECTIONS	
DELTA	[SCHNEIDER]
+24V	+24V
MI1	LI1
MI3	—
ACM	COM
DCM	+24V
AVI	AI1
RA	RIA
RC	RIC



Use this electrical diagram for coolant pumps that are being driven by the Variable Frequency Drive With Cable 33-9334C.

33-9334C VFD Detail Cable Diagram